Jonghoon Kwon

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Professional Experience

05.2024–Present **Senior Scientist**, *ETH Zürich*, Switzerland.
- Networking for AI, AI for networking
  - Convergence-aware optimization of distributed AI training across data centres
  - ML-based prediction of path dynamics and intent-driven path selection
  - Trustworthy AI infrastructure: sovereign, verifiable networks for federated AI
- SCIERA: SCION Education, Research and Academic Network
  - Develop research collaboration with national research and education networks (NRENs)
  - Establish a global ring connection circulating Asia, Europe, and North America
  - Expand deployment map to Middle East, Africa, and South America
- Carbon-aware Global Routing in Path-aware Networks
  - Estimate and disseminate carbon intensity of inter-domain paths
  - Enable carbon-aware path selection, reducing the carbon intensity of inter-domain traffic
- Programmable Network Telemetry
  - Next-generation network debugging infrastructure
  - Distributed data-plane operations from multiple vantage points
  - Segment-by-segment network debugging capability

05.2020–04.2024 **Senior Assistant (Established Researcher I&II)**, *ETH Zürich*, Switzerland.
- Secure Internet Backbone
  - provide origin security to ASes without RPKI
- Flexible Network Zoning at Inter-domain Scale
  - Develop a method to facilitate secure and flexible zone transfer across autonomous systems relying on different intra-domain virtualization protocols (e.g., Trill, SPB, VLAN, and VXLAN)
- Global Time Synchronization for Future Internet
  - Investigate the level of time synchronization that can be achieved on a global scale
  - Secure time synchronization to prevent attackers from de-synchronizing hosts and autonomous systems
- Secure Inter-domain Network Virtualization
  - Motivated from the lack of security property in the modern network virtualization techniques
  - Proposed a network virtualization architecture that ensures security, scalability and flexibility
- Dynamically Recreatable Key Creation with Quantum Mechanism
  - Collaborate with ID Quantique SA on developing a prototype for dynamic key creation mechanism that leverages quantum key distribution (QKD) and quantum random number generator (QRNG).
  - Integrate future Internet architecture and quantum mechanisms to achieve secure key exchanges amongst autonomous systems

01.2017–04.2020 **Postdoctoral Researcher**, *ETH Zürich*, Switzerland.
- SCIONLab: A Next-Generation Internet Testbed
  - Construct a global network testbed distributed across Europe, North America, Asia and Australia
  - Construct a testbed backbone that provides 20Gbps global ring connectivity and 5 different point-of-presents (i.e., Amsterdam, Seattle, Chicago, Daejeon, and Hongkong)
  - Developed a centralized management system and an Web interface that supports client-driven network participant
  - `https://www.scionlab.org`

09.2016–12.2017 **Research Staff**, *Korea University*, South Korea.
- Worked with John Milburn on a fast invocation of replica servers in cloud for DDoS defense

09.2012–03.2013 **Research Intern**, *Microsoft Research Asia*, China.
- Worked with Dr. Bin Zhu on developing an identity inference model for social media services

Education

2011–2016 **Ph.D., Computer Science and Engineering**, *Korea University*, South Korea.
- Dissertation: A Scalable Botnet Countermeasures for Large-scale DNS Traffic
- *Committee*: Heejo Lee, Saewoong Bahk, Hyogon Kim, Hoh Peter In, Junbeom Hur

2008–2010 **M.S., Computer Science and Engineering**, *Korea University*, South Korea.

2000–2007 **B.S., Computer Science**, *Korea University*, South Korea.
- 3 years of leave of absense due to military service (2002–2004)

Awards

Best Paper Award Nomination, ACM International Conference on Future Energy Systems (e-Energy), 2023

Best Paper Award, ACM Conference on emerging Networking EXperiments and Technologies (CoNEXT), 2021

Best Paper Award, IEEE International Conference on Network Protocols (ICNP), 2020

Best Paper Award, International Conference on Applications and Techniques in Information Security, 2014

Conference Publications

CoNEXT'26 F. Stöger, J. van Bommel, **J. Kwon**, P. Bamert, S. Boss, A. Eggerth, and A. Perrig,
Authenticated Browsing: Reconciling Application-Layer Flooding Defense with Data Confidentiality,
ACM Conference on Emerging Networking Experiments and Technologies (CoNEXT), Dec. 2026.

IEEE ICNP'26 Y. Lin, C. Mark, **J. Kwon**, and A. Perrig,
TRACER: Flexible and Responsible Distributed Active Network Measurements,
IEEE International Conference on Network Protocols, Oct. 2026.

IEEE ICDCS'26 E. E. Moghadam, M. Wyss, **J. Kwon**, M. Frei, Y. Hu, A. Perrig, and A. Sonnino,
Signet: Scalable Network-Driven Proof of Notification for Blockchain Systems,
IEEE International Conference on Distributed Computing Systems, Jun. 2026.

ACM SIGCOMM'25 F. Wirz, M. Gartner, J. v. Bommel, E. E. Moghadam, G. H. Cimaszewski, A. He, Y. Zhang, H. Birge-Lee, F. Kottmann, C. Krähenbühl, **J. Kwon**, K. Mavromati, L. Wang, D. Bertolo, M. Canini, B. Cho, R. A. Ferreira, S. P. Green, D. Hausheer, J. Hur, X. Jia, H. Lee, P. Mittal, O. Oaiya, C. Park, A. Perrig, J. Sobieski, Y. Sun, C. Wang, and K. Wierenga,
Scaling SCIERA: A Journey Through the Deployment of a Next-Generation Network,
ACM SIGCOMM, Sep. 2025.

IEEE S&P'25 M. Wyss, M. Frei, **J. Kwon**, A. Perrig,
Low-cost and Robust Global Time Synchronization,
IEEE Symposium on Sercurity and Privacy, May. 2025.

IEEE S&P'25 M. Frei, **J. Kwon**, M. Wyss, A. Perrig,
Poster: Clockwire – Multi-path Time Synchronization Made Practical,
IEEE Symposium on Sercurity and Privacy, May. 2025.

IEEE ICC'25 **J. Kwon**, J. Modanese, J. Nieto, A. Perrig,
Path-aware Access Control: Granting Access with Transit Network Attributes,
IEEE International Conference on Communications, Jun. 2025.

ICDCS'24 S. Tabaeiaghdaei, F. Cost, **J. Kwon**, P. Bamert, Y. Hu, A. Perrig,
Debuglet: Programmable and Verifiable Inter-domain Network Telemetry,
IEEE International Conference on Distributed Computing Systems (ICDCS), Jul. 2024.

USENIX SEC'23 **J. Kwon**, J. Song, J. Hur, A. Perrig,
Did the Shark Eat the Watchdog in the NTP Pool? Deceiving the NTP Pool's Monitoring System,
USENIX Security Symposium 2023, Aug. 2023.

ACM e-Energy'23 S. Tabaeiaghdaei, S. Scherrer, **J. Kwon**, A. Perrig,
Carbon-Intelligent Global Routing in Path-Aware Networks,
ACM International Conference on Future Energy Systems (e-Energy), Jun. 2023.

ACM SACMAT'23 S. Mitani, **J. Kwon**, N. Ghate, T. Singh, H. Ueda, A. Perrig,
Qualitative Intention-aware Attribute-based Access Control Policy Refinement,
ACM Symposium on Access Control Models and Technologies (SACMAT), Jun. 2023.

USENIX SEC'22 H. Birge-Lee, J. Wanner, G. Cimaszewski, **J. Kwon**, L. Wang, F. Wirz, P. Mittal, A. Perrig, Y. Sun.,
Creating a Secure Underlay for the Internet,
USENIX Security Symposium 2022, Aug. 2022.
Acceptance rate: 18%

SRDS'22 M. Frei, **J. Kwon**, S. Tabaeiaghdaei, M. Wyss, C. Lenzen, A. Perrig,
G-SINC: Global Synchronization Infrastructure for Network Clocks,
Symposium on Reliable Distributed Systems (SRDS), Sep. 2022.
Acceptance rate: 23%

ICDCS'22 M. Blarer, **J. Kwon**, V. G. Narbel, A. Perrig,
Consent Routing: Towards Bilaterally Trusted Communication Paths,
IEEE International Conference on Distributed Computing Systems (ICDCS), Jul. 2022.
Acceptance rate: 19.9%

DSN'22 C. Koh, **J. Kwon**, J. Hur,
BLAP: Bluetooth Link Key Extraction and Page Blocking Attacks,
IEEE/IFIP International Conference on Dependable Systems and Networks (DSN), Jun. 2022.
Acceptance rate: 18.7%

CoNEXT'21 C. Krähenbühl, S. Tabaeiaghdaei, C. Gloor, **J. Kwon**, D. Hausheer, A. Perrig, D. Roos,
Deployment and Scalability of an Inter-Domain Multi-Path Routing Infrastructure,
ACM Conference on emerging Networking EXperiments and Technologies (CoNEXT), Dec. 2021.
Acceptance rate: 15.8%
Best Paper Award & Best Presentation Award

K. Gärtner, **J. Kwon**, D. Hausheer,
SpeedCam: Towards Efficient Flow Monitoring for Multipath Communication,
IEEE/IFIP International Workshop on Analytics for Network and Service Management, May. 2021

NDSS'21 **J. Kwon**, C. Hähni, P. Bamert, A. Perrig,
MONDRIAN: Comprehensive Inter-domain Network Zoning Architecture,
Symposium on Network and Distributed System Security (NDSS), Feb. 2021.
Acceptance rate: $< 16\%$

ICNP'20 **J. Kwon**, J. García-Pardo, M. Legner, F. Wirz, M. Frei, D. Hausheer, A. Perrig,
SCIONLab: A Next-Generation Internet Testbed,
IEEE International Conference on Network Protocols (ICNP), Oct. 2020.
Acceptance rate: 16.3%
Best Paper Award

NDSS'20 **J. Kwon**, T. Lee, C. Hähni, A. Perrig,
SVLAN: Secure & Scalable Network Virtualization,
Symposium on Network and Distributed System Security (NDSS), Feb. 2020.
Acceptance rate: 17.4%

ATIS'14 H. Li, H. Kwon, **J. Kwon**, H. Lee,
A Scalable Approach for Vulnerability Discovery Based on Security Patches,
Applications and Techniques in Information Security, pp. 109-122, Nov. 2014.
Best Paper Award

CNS'14 **J. Kwon**, J. Jeong, J. Lee, H. Lee,
DroidGraph: Discovering Android Malware by Analyzing Semantic Behavior,
IEEE Conference on Communications and Network Security, pp. 345-346, Oct. 2014

MALWARE'14 **J. Kwon**, J. Kim, J. Lee, H. Lee, A. Perrig,
PsyBoG: Power Spectral Density Analysis for Detecting Botnet Groups,
IEEE Conference on Malicious and Unwanted Software, pp. 85-92, Oct. 2014

J. Jeong, D. Seo, C. Lee, **J. Kwon**, H. Lee, J. Milburn,
MysteryChecker: Unpredictable Attestation to Detect Repackaged Malicious Applications in Android
IEEE Conference on Malicious and Unwanted Software, pp. 50-57, Oct. 2014

MALWARE'12 **J. Kwon**, H. Lee,
BinGraph: Discovering Mutant Malware using Hierarchical Semantic Signature,
IEEE Conference on Malicious and Unwanted Software, Oct. 2012

J. Kwon, J. Lee, H. Lee,
Hidden Bot Detection by Tracing Non-human Generated Traffic at the Zombie Host,
Information Security Practice and Experience Conference, May. 2011

NPSEC'10 J. Lee, **J. Kwon**, H. Shin, H. Lee,
Tracking Multiple C&C Botnets by Analyzing DNS Traffic,
Workshop on Secure Network Protocols, pp. 67-72, Oct. 2010.

IFIP SEC'08 J. Han, **J. Kwon**, H. Lee,
HoneyID : Unveiling Hidden Spywares by Generating Bogus Events,
IFIP Information Security Conference (SEC), Vol. 278, pp. 669-673, Sep. 2008

Journal Publications

J. Kwon, J. Lee, H. Lee, A. Perrig,
PsyBoG: A scalable botnet detection method for large-scale DNS traffic,
Computer Networks, Vol. 97, pp. 48-73, Mar. 14. 2016

H. Lee, H. Kwon, **J. Kwon**, H. Lee,
CLORIFI: Software Vulnerability Discovery using Code Clone Verification,
Concurrency and Computation: Practice and Experience, Vol. 28, No. 6, pp. 1900-1917, Apr. 14. 2015

J. Kwon, D. Seo, M. Kwon, H. Lee, A. Perrig, H. Kim,
An incrementally deployable anti-spoofing mechanism for software-defined networks,
Computer Communications, Vol. 64, pp. 1-20, Jun. 15. 2015

Book

The Complete Guide to SCION, *Springer*, 2022.
- Chapter 8. Extensions for the Control Plane
- Chapter 13. Deployment and Operation
- Chapter 14. SCIONLab Research Testbed

Patents

H. Lee, H. Li, **J. Kwon**, H. Kwon,
Software Vulnerability Analysis Method and Device,
International, Registration, US9946880, Apr. 2018

H. Lee, **J. Kwon**, J. Kim,
System and Method for Detecting Malicious Code in Document Files,
Domestic, Registration, 10-1641295, Jul. 2016

H. Lee, H. Li, H. Kwon, **J. Kwon**,
Analysis Device and Method for Software Security,
Domestic, Registration, 10-1568224, Nov. 2015

H. Lee, **J. Kwon**, J. Lee, J. Lee, T. Kim, H. Jeong, C. Im, S. Ji, J. Oh, D. Kang,
System and Method for Detecting Malicious Code,
Domestic, Registration, 10-1230271, Jan. 2013

H. Lee, **J. Kwon**, J. Lee,
Method and Apparatus for Detecting Bot Process,
Domestic, Registration, 10-1158464, Jun. 2012

H. Lee, J. Lee, **J. Kwon**,
Method and System for Detecting Botnets using Domain Name Service Queries,
Domestic, Registration, 10-1182793, Sep. 2012

Technical Reports

H. Choi, **J. Kwon**, I. Kim, H. Lee,
Botnet Attacks using Malicious Codes,
Business and Computer, pp. 144-147, Jul. 2008.

Teaching Experience

ETH Zürich, *Dept. of Computer Science*, Institute for Information Security.
- Computer Networks (Spring 2024, Spring 2025)
- Current Topics in Information Security (Fall 2019, Fall 2020, Fall 2021, Fall 2022, Fall 2023)

Korea University, *Dept. of Computer Science and Engineering*.
- Information Security (Spring 2011)
- Software Security (Fall 2009)
- Theory of Computation (Spring 2008, Fall 2011)

Professional Service

2023-Present **ACM SIGCOMM Workshop on Future of Internet Routing & Addressing (FIRA)**, Technical Program Committee,

2022-Present **KIAT Europe**, Korea Technology Advisory Group (K-TAG)

2021 **ACM SIGCOMM Workshop on Technologies, Applications, and Uses of a Responsible Internet (TAURIN)**, Technical Program Committee

Industrial Collaboration

2023-Present **Mysten Labs**, *USA*, Topic: Secure Blockchain Network.

2022-Present **Intercloud SAS**, *Paris, France*, Topic: Secure & Reliable Inter Cloud Connectivity.

2022-Present **SWITCH**, *Zürich, Switzerland*, Topic: Secure Education Network.

2021-Present **KISTI**, *Daejoen, Korea*, Topic: ScienceDMZ, Data Exchange Node.

2020-2023 **NEC**, *Yokohama, Japan*, Topic: 5G Security, Path-based Access Control.

2020-Present **LGU+**, *Seoul, Korea*, Topic: Secure Inter-domain Connection b/w Korea and Europe.

2020-Present **Zürcher Kantonal Bank**, *Zürich, Switzerland*, Topic: Secure Banking Network.

2019-2020 **ID Quantique**, *Geneva, Switzerland*, Topic: Inter-Datacenter Quantum Key Distribution.

2019-Present **Anapaya Systems AS**, *Zürich, Switzerland*, Topic: Hidden Path, Global Time Synchronization.

2022-Present **Labrador Labs (IoTCube)**, *Seoul, Korea*, Topic: SCION Deployment in Korea.

Skills

O.S. Windows, Ubuntu Linux, OSX

P.L. C/C++, C#, Python, Golang, ML

Tools Coordination tool (ZooKeeper, Ansible), Virtualization tool (Docker), Database (MySQL), Revision control tool (Git, SVN), Machine learning tool (Weka), Software-defined networking simulator (Mininet), Network simulator (Qualnet), Reversing tool (OllyDbg), Graph processing (Pajek), LaTeX, GNU plot, Wireshark

Research Interests

Future Internet Architecture, Software Defined Networking, Network Virtualization, Malware Analsys and Detection, Software Vulnerability Detection

References

Adrian Perrig,
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Heejo Lee,
Professor, Korea University, Seoul, South Korea
heejo@korea.ac.kr, +82 2 3290 3208

David Hausheer,
Professor, Otto-von-Guericke-University, Magdeburg, Germany
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(more references available upon request)